Immunology

Ingyu Bahng

September 2026

Contents

1 Complement System

1.1 MAC Complex Attack

Core Concept 1.1: Complement System

The **complement system** is an enzymatic cascade of more than 30 soluble plasma proteins and membrane receptors that function as a core branch of **innate immunity** while also amplifying the effector functions of adaptive antibodies.

Most circulating complement proteins are synthesized predominantly by **hepatocytes** (*liver cells*) and released as **inactive** precursors, although several immune and stromal cell populations also produce complement components locally.

Core Concept 1.2: MAC Complex Attack

The **membrane attack complex (MAC)** is the shared **terminal** effector of complement activation. The **classical** (*Core Concept 1.3*), **lectin** (*Core Concept 1.4*), and **alternative** (*Core Concept 1.5*) pathways all converge on the generation of a **C5 convertase**, whose cleavage of C5 initiates assembly of the terminal complement components.

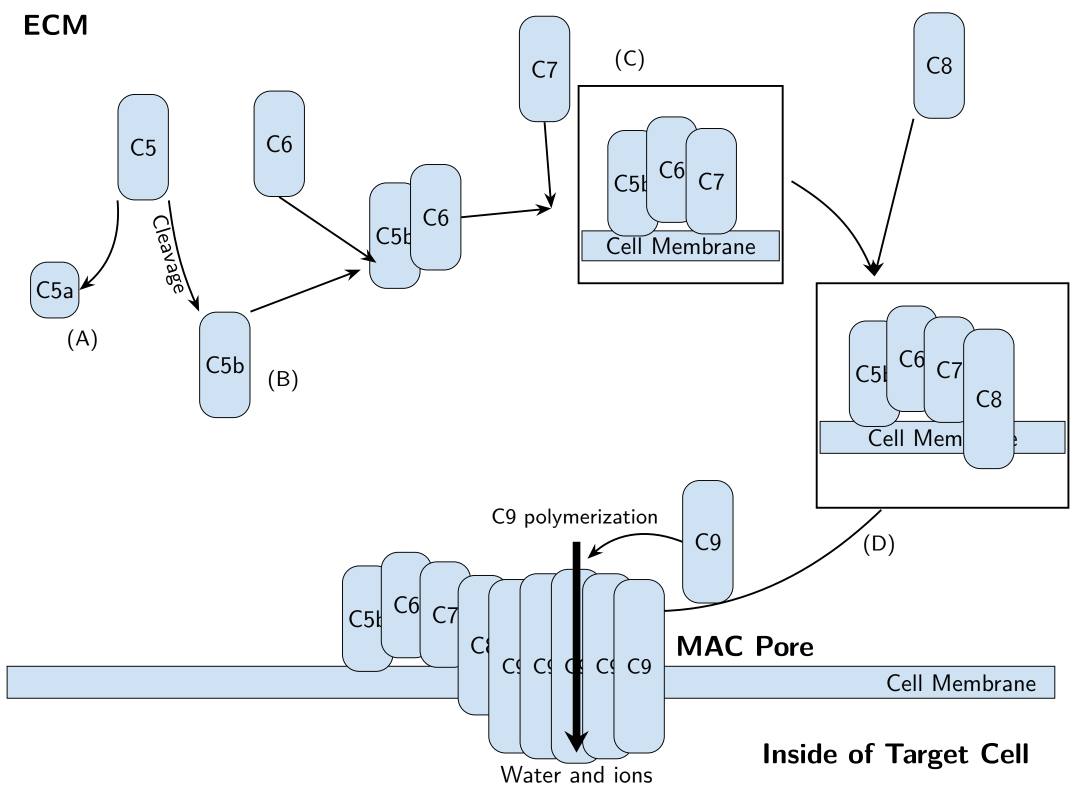

Figure 1: Membrane attack complex (MAC); *all different complement pathways converge to this pathway*; (A) *C5a acts as a strong* ***anaphylatoxin*** *(Core Concept 1.6)*; (B) *C5b is unstable when left unbound*; (C) *C7 recruitment exposes a hydrophobic site that anchors the complex to the membrane*; (D) *C8 recruitment drives partial insertion of the complex into the lipid bilayer*

As shown in Figure 1, the completed MAC achieves **membrane lysis** by forming a transmembrane pore composed of polymerized C9 subunits; structural studies generally describe a pore containing approximately

18 C9 subunits. This pore permits the **uncontrolled movement of water and ions across the membrane**, disrupting ionic homeostasis and potentially inducing osmotic rupture. Several types of susceptible targets can be damaged in this way, including the following.

- **Enveloped viruses**, whose host-derived lipid membranes may be disrupted by terminal complement deposition
- **Gram-negative bacteria** (*FIGURE REFEERNCE FROM UNIT 1*), whose outer membrane is accessible to MAC insertion, in contrast to the thick peptidoglycan wall that generally protects *Gram-positive bacteria* from direct MAC-mediated lysis
- Susceptible **eukaryotic parasites**
- **Erythrocytes** (*red blood cells*) under pathological conditions in which complement regulation fails or inappropriate activation occurs
- Antibody-targeted **cancer cells**, provided that complement activation overcomes the cells' membrane-bound complement regulators.

1.2 Complement Pathways

Core Concept 1.3: Classical Pathway

The **classical pathway** is the antibody-dependent route of complement activation, triggered when the C1 complex engages antigen-bound immunoglobulin.

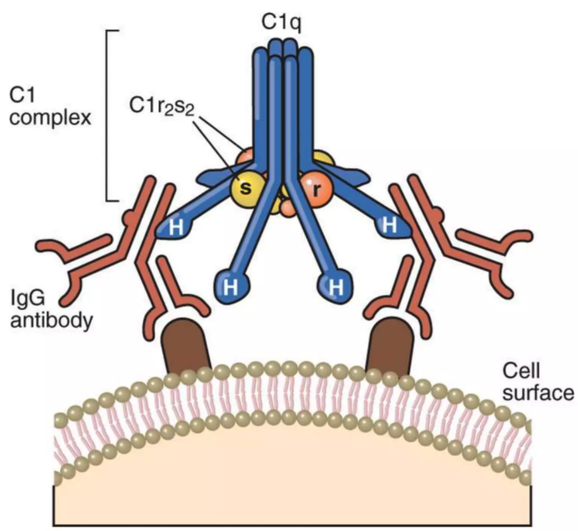

Figure 2: C1 complex attached to the Fc stem of an IgG bound to an antigen

As shown in Figure 2, the inactive C1 complex is composed of one hexameric C1q recognition molecule bound to two C1r and two C1s serine proteases. Each of C1q's six arms terminates in a globular head assembled from C1qA, C1qB, and C1qC chains (*not shown*). C1q recognizes sites within the Fc region of antigen-bound immunoglobulin, and the local Fc conformation and associated glycans influence the accessibility and efficiency of this interaction. **Engagement of multiple Fc regions** by C1q induces a conformational change that activates C1r, which in turn activates C1s, initiating proteolytic cleavage of C4 and C2.

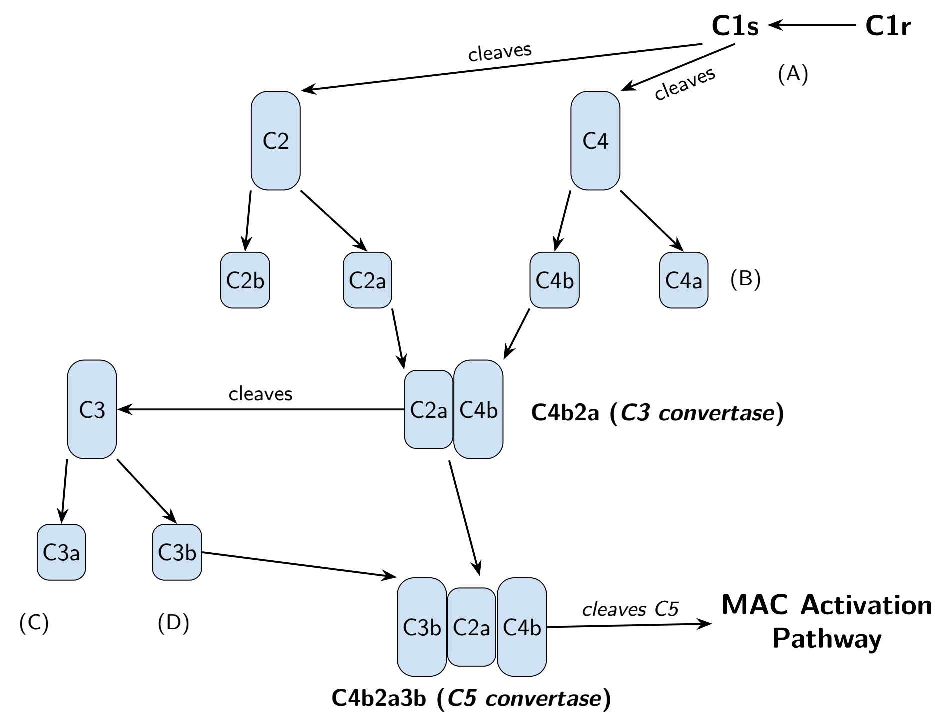

Figure 3: Classical complement pathway activation cascade; (A) *activated C1r cleaves C1s, producing an active protease*; (B) *C4a is a weak* ***anaphylatoxin*** *(Core Concept 1.6)*; (C) C3a goes onto act as an **anaphylatoxin**; (D) *C3b also functions as an* ***opsonin*** *(Core Concept 1.6), coating targets for phagocytic recognition*

It's important to note that complement-activating antibody isotypes differ substantially in their ability to recruit C1q.

- **IgM** is a **highly efficient** activator. Soluble pentameric IgM generally adopts a planar conformation, whereas when bound to antigen, the pentamer takes up a **staple-like conformation** (*FIG REFERENCE*) that exposes C1q-binding sites in its Fc regions.

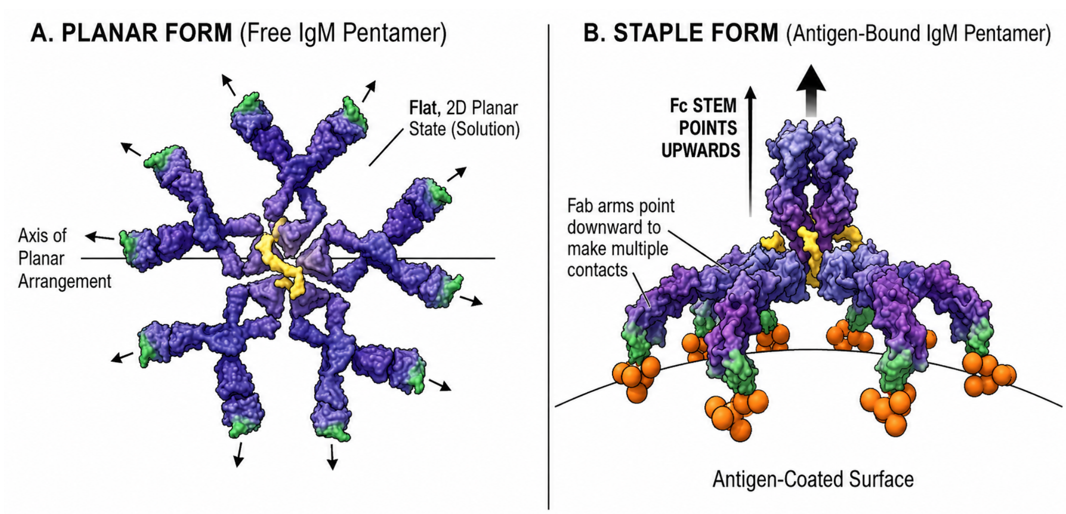

Figure 4: Antigen binding induces a conformational shift in pentameric IgM, from a planar resting state (*left*) to a staple-like conformation (*right*) that exposes C1q-binding sites in the Fc stems

- Because **IgG** is monomeric, efficient C1 activation normally requires several antigen-bound IgG molecules to cluster their Fc regions and provide multivalent engagement of C1q. This requirement limits inappropriate activation by freely circulating IgG.

  Among human IgG subclasses, classical pathway activation potency varies considerably: IgG3, owing to its comparatively long hinge region, is the most efficient activator, followed by IgG1, while IgG2 is relatively weak and IgG4 does not appreciably activate the classical complement pathway.
- **IgA, IgE, and IgD** do not ordinarily initiate the classical complement pathway.

Core Concept 1.4: Lectin Pathway

The **lectin pathway** parallels the classical pathway downstream of its initial recognition event. Instead of using C1q to recognize antigen-bound antibody, it begins when soluble **PRR** molecules, namely **mannose-binding lectin (MBL)**, bind characteristic carbohydrate structures on a microbial surface and activate associated serine proteases called **MASP**.

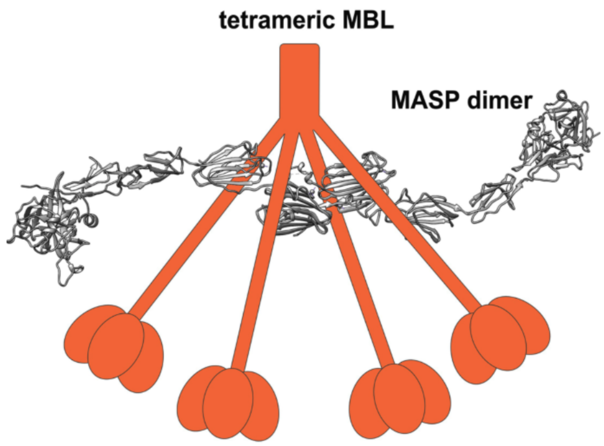

Figure 5: Structure of tetrameric MBL bound to a MASP dimer

In comparison to the **classical pathway** (*Core Concept 1.3*), MBL resembles C1q in that it occurs in clusters. Additionally MASP molecules are anlogous to C1r and C1s

Lectin Pathway	**Classical Pathway**
Initiated by mannose and other carbohydrates	Initiated by antibodies, using the domain with the oligosaccharide
Changes conformation of MBL	Changes conformation of C1q
Activates MASP	Activates C1r/C1s

Table 1: Comparison of the lectin and classical complement pathways

Following activation of the lectin-pathway proteases, cleavage of C4 and C2 produces the same C3 convertase, C4b2a, that functions in the classical pathway. The two pathways are therefore identical from C3-convertase formation onward.

Core Concept 1.5: Alternative Pathway

The **alternative pathway** is an evolutionarily ancient route of complement activation that relies on innate pattern recognition of antigens present on pathogenic surfaces, including teichoic acid (*bacterial*), zymosan (*fungal*), certain tumor-associated compounds, and trypanosome (*unicellular eukaryote*) surface markers. Like the classical (*Core Concept 1.3*) and lectin (*Core Concept 1.4*) pathways, the alternative pathway ultimately converges on the formation of a C5 convertase, which cleaves C5 and initiates assembly of the MAC, as outlined in Core Concept 1.2.

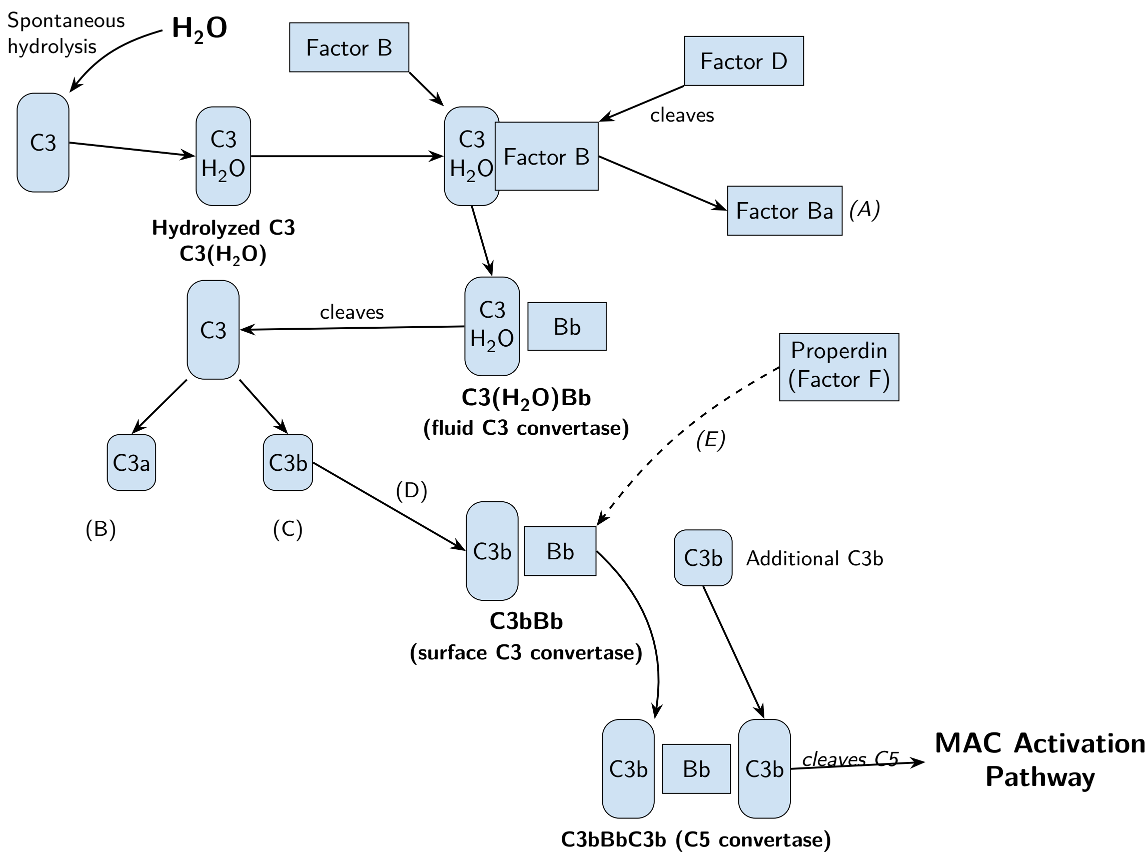

Figure 6: Alternative complement pathway activation cascade; (A) *factor Ba diffuses away*; (B) *C3a goes onto act as an* ***anaphylatoxin*** *(Core Concept 1.6)*; (C) *C3b also serves as an* ***opsonin*** *(Core Concept 1.6), coating activating surfaces*; (D) *factor B binding and factor D cleavage of factor B into Bb*; (E) *properdin—sometimes called factor F— binds and stabilizes C3bBb on the activating surface*

Core Concept 1.6: Side Signal Pathways & Other

Complement activation produces several effects in addition to MAC-mediated lysis. Cleavage products generated throughout the cascade promote inflammation, coat targets for immune recognition, neutralize pathogens, and facilitate the removal of immune complexes.

An **anaphylatoxin** is a small peptide fragment released as a byproduct of proteolytic cleavage during the complement cascade; upon binding specific receptors on immune and vascular cells, these fragments trigger localized inflammatory responses. C3a, C4a, and C5a are all anaphylatoxins, arising respectively from cleavage of C3, C4, and C5 during the unique complement and converged MAC pathways.

- **C3a**: C3a stimulates degranulation of **mast cells** and **basophils**, promoting the release of **histamine** and other inflammatory mediators that increase vascular permeability.

- **C4a**: C4a exhibits **comparatively weak anaphylatoxin activity** relative to C3a and C5a, but nonetheless contributes to local inflammatory signaling following its release during C4 cleavage.
- **C5a**: Among the anaphylatoxins, C5a is the most potent, functioning as a chemoattractant that recruits and activates **neutrophils**, **monocytes**, and other **myeloid cells** at sites of complement activation.

In addition to their inflammatory functions, complement fragments deposited on target surfaces mark pathogens and immune complexes for recognition, neutralization, and clearance by phagocytic mechanisms.

(a) **Opsonization**: Covalently deposited complement fragments, particularly C3b and iC3b, coat microbial surfaces and enhance their recognition and phagocytosis through complement receptors.

(b) **Viral Neutralization**: Complement deposition can sterically hinder viral attachment or entry, promote virion aggregation, and facilitate clearance by phagocytes.

(c) **Immune-Complex Clearance**: Complement-tagged antigen–antibody complexes bind complement receptor 1 on erythrocytes and are transported through the circulation to the liver and spleen, where phagocytes remove and degrade them.

2 Major Histocompatibility Complex

2.1 MHC Structure & Genetics

Core Concept 2.1: MHC Characteristics

<FIGURE>

<TABLE>

Core Concept 2.2: MHC Gene Sequences

<FIGURE>

The human major histocompatibility complex is encoded within a gene-dense region on the short arm of chromosome 6.

The linked set of MHC alleles inherited together on a single chromosome constitutes an **MHC haplotype**.

Human MHC molecules are designated **human leukocyte antigens (HLAs)**. Their extracellular domains adopt immunoglobulin-superfamily folds, although the MHC region also contains many genes whose products are not immunoglobulin-superfamily members.

Each diploid individual inherits one MHC haplotype maternally and one paternally, and both sets of alleles are expressed codominantly.

(a) **MHC I**

Classical MHC class I molecules are encoded by the **HLA-A, HLA-B, and HLA-C** loci, whose allelic products differ in peptide-binding specificity.

The HLA-A, HLA-B, and HLA-C loci are exceptionally polymorphic, and modern allele databases contain many thousands of named variants; the historical counts of 68, 125, and 44 variants, respectively, are therefore obsolete underestimates.

An individual may express as many as six distinct classical MHC class I allotypes when the maternal and paternal alleles at all three loci differ.

The HLA-A, HLA-B, and HLA-C genes encode only the polymorphic α chains. The invariant **β2-microglobulin** component is encoded separately by the *B2M* gene on chromosome 15.

These classical class I loci exhibit extraordinary allelic polymorphism, particularly within sequences that encode the peptide-binding groove.

The diversity of classical MHC class I molecules among individuals primarily reflects germline polymorphism rather than somatic DNA rearrangement.

(b) **MHC III**

The MHC class III region does not encode the peptide-presenting molecules characteristic of MHC classes I and II.

It is designated part of the MHC because of its physical location between the class I and class II regions on chromosome 6.

This region contains genes encoding several immune mediators, including complement components, tumor necrosis factors, and heat-shock proteins.

(c) **MHC II**

The classical MHC class II region contains the **HLA-DP, HLA-DQ, and HLA-DR** loci. Each class II molecule is an αβ heterodimer, and both chains contribute to the peptide-binding groove.

Codominant expression of maternal and paternal class II genes, together with permitted pairing between certain α and β chains, allows an individual to display a diverse set of MHC class II molecules; the exact number depends on the inherited haplotypes and the expressed DRB genes rather than a universal maximum of twelve.

Classical MHC class II loci are highly polymorphic, with much of this variation concentrated in residues that shape the peptide-binding groove.

As with class I molecules, interindividual class II diversity primarily reflects inherited germline polymorphism rather than somatic DNA rearrangement.

(d) **Other**

The class II region also contains genes encoding components of the MHC class I antigen-processing pathway, including *TAP1*, *TAP2*, *PSMB8*, and *PSMB9*.

Nonclassical MHC class I molecules perform more specialized functions in immune recognition, molecular transport, and antigen presentation.

Core Concept 2.3: MHC Mendalian Genetics

(a) **Non-Recombinant Inheritance**

<FIGURE>

Because the MHC loci are tightly linked, their alleles are usually inherited together as a haplotype. In the absence of recombination, two full siblings have a 25% probability of inheriting the same pair of parental MHC haplotypes.

More generally, the term **haplotype** denotes a set of linked alleles or sequence variants inherited together from one parent; its use is not restricted to the MHC.

(b) **Recombinant Inheritance**

<FIGURE>

Meiotic crossing over within the MHC can generate a recombinant haplotype that combines portions of the parental class I and class II regions. Such recombination reduces the probability that siblings will match across the entire MHC.

Over evolutionary time, recombination contributes additional haplotypic diversity to the population.

Inbreeding increases the probability that an individual will inherit identical or closely related MHC haplotypes.

In a highly inbred population, limited introduction of new haplotypes increases MHC homozygosity, such that an individual may carry the same haplotype on both homologous chromosomes.

An individual homozygous across the classical MHC loci expresses fewer distinct allotypes than a fully heterozygous individual—typically one product from each of HLA-A, HLA-B, and HLA-C, together with a correspondingly reduced class II repertoire.

2.2 MHC Antigen Preparation

Core Concept 2.4: Endogenous MHC I (T_c Cells)

(1) **Proteasomal Degradation of Antigen to Peptide**

<FIGURE>

Cytosolic proteins destined for degradation are commonly tagged with chains of the small protein **ubiquitin**, which direct them to the proteasome for ATP-dependent unfolding and proteolysis.

The proteolytic 20S core of the proteasome consists of four stacked heptameric rings. Cellular stress and inflammatory signals can increase protein turnover and alter proteasome composition, thereby shaping the pool of peptides available for MHC class I presentation.

During inflammation, interferon signaling promotes incorporation of the inducible catalytic subunits LMP2 and LMP7 into the **immunoproteasome**. These substitutions favor the production of peptides with termini suitable for MHC class I binding, although the products may require additional trimming and are not uniformly nine residues long.

Additional cytosolic peptidases and alternative proteolytic pathways can generate or trim antigenic peptides when conventional proteasomal processing is insufficient.

LMP2 and LMP7 are encoded by the *PSMB9* and *PSMB8* genes, respectively, which lie within the MHC class II region and exhibit genetic polymorphism.

(2) **Transport to the RER**

<FIGURE>

The transporter associated with antigen processing (TAP), a heterodimer of TAP1 and TAP2, binds peptides on the cytosolic face of the rough endoplasmic reticulum (RER) and uses ATP hydrolysis to translocate them into the RER lumen.

The polymorphic *TAP1* and *TAP2* genes are located within the MHC class II region.

TAP structure reflects its function as an ATP-binding cassette transporter.

- TAP is an obligate heterodimer composed of TAP1 and TAP2.
- Its peptide-binding site is accessible from the cytosol.
- Cytosolic nucleotide-binding domains hydrolyze ATP to power peptide translocation.
- Each subunit contains multiple hydrophobic transmembrane helices that together form the peptide-conducting pathway through the RER membrane.

(3) **Assembly of MHC I**

(a) *Calnexin-α to α-β*

<FIGURE>

The newly synthesized MHC class I α chain is inserted into the RER membrane, while soluble β2-microglobulin enters the RER lumen.

The membrane-bound chaperone calnexin associates with the α chain and promotes its proper folding.

Binding of β2-microglobulin stabilizes the α chain and permits calnexin to dissociate.

(b) *α-β-Calreticulin-Tapasin-TAP-ERp57*

<FIGURE>

The partially assembled MHC class I heterodimer joins calreticulin, tapasin, and ERp57 to form the peptide-loading complex associated with TAP.

Calreticulin and ERp57 stabilize the peptide-receptive MHC class I molecule and support quality control during peptide loading.

Tapasin bridges MHC class I to TAP, positioning the empty peptide-binding groove near the source of newly translocated peptides.

(c) *α-β-Calreticulin-Tapasin-TAP-Peptide*

<FIGURE>

Peptides released by TAP sample the MHC class I groove, while the peptide-loading complex favors the selection of high-affinity cargo. Stable peptide binding induces conformational maturation and ultimately releases the loaded MHC class I molecule from its chaperones.

(d) *α-β-Peptide to Golgi*

<FIGURE>

After calreticulin and tapasin dissociate, the stable peptide–MHC class I complex exits the RER and proceeds through the Golgi apparatus.

(4) **Transportation to Exterior**

<FIGURE>

Following processing and sorting in the Golgi, vesicles deliver peptide–MHC class I complexes to the plasma membrane by exocytosis, exposing the peptide-binding platform to surveillance by CD8$^+$ T cells.

Core Concept 2.5: Exogenous MHC I / Cross Presentation (T_c Cells)

(1) **Antigen Uptake**

<FIGURE>

Specialized antigen-presenting cells, particularly dendritic cells, internalize exogenous antigens through endocytosis or phagocytosis.

(2) **Proteasomal Degradation of Antigen to Peptide**

<FIGURE>

Internalized proteins are processed through either a cytosolic route, in which antigen reaches the cytosol for proteasomal degradation and TAP-dependent loading, or a vacuolar route, in which processing and class I loading occur within endocytic compartments. The resulting peptide–MHC class I complexes are transported to the cell surface.

Cross-presentation is performed most efficiently by specialized dendritic-cell subsets. During **cross-priming**, these cells provide peptide–MHC class I together with co-stimulatory and cytokine signals that activate naive CD8$^+$ T cells; presentation in the absence of appropriate co-stimulation may instead promote tolerance.

Cross-presentation enables CD8$^+$ T-cell responses against viruses or tumors that do not directly infect professional antigen-presenting cells, thereby allowing timely cytotoxic immunity to cell-associated exogenous antigens.

Core Concept 2.6: Exogenous MHC II

(1) **Antigen Processing**

(a) *Antigen Uptake*

<FIGURE>

Exogenous antigens enter antigen-presenting cells through receptor-mediated endocytosis, macropinocytosis, or phagocytosis. Phagocytosis is specialized for the engulfment of large particles, including intact microorganisms.

Endocytosed material enters an endosome, whereas phagocytosed material is initially enclosed within a phagosome.

(b) *Hydrolysis in Vesicles*

<FIGURE>

As endocytic compartments mature, progressive acidification activates resident hydrolases that degrade internalized proteins and remove associated carbohydrates and lipids. Proteolysis generates peptides of variable length, commonly approximately 13–18 residues, that can bind the open-ended groove of MHC class II.

- Early endosomes maintain a mildly acidic pH of approximately 6.0–6.5.
- Late endosomes and endolysosomes acidify further, generally reaching approximately pH 5.0–6.0.
- Lysosomes and phagolysosomes provide the most acidic and hydrolytic environment, typically near pH 4.5–5.0.

(2) **MHC II**

(a) *Preparation of MHC II*

<FIGURE>

Following cotranslational insertion of the MHC class II α and β chains into the RER membrane, the heterodimer associates with the invariant chain (Ii; CD74).

Although a simplified diagram may depict a single invariant chain, Ii trimerizes and can associate with three MHC class II $\alpha\beta$ heterodimers.

The invariant chain occupies the peptide-binding groove to prevent premature loading of RER-derived peptides, stabilizes the nascent heterodimer, promotes exit from the RER, and contains sorting signals that direct the complex into the endosomal pathway.

(b) *Transportation of Golgi*

<FIGURE>

The MHC class II–Ii complex traverses the Golgi and is subsequently sorted into acidified endosomal compartments containing processed exogenous antigen.

(3) **Fusion with Endocytic Vesicles/Phagolysosomes**

<FIGURE>

After leaving the Golgi, vesicles containing MHC class II–Ii complexes enter the endocytic pathway and mature toward late endosomal peptide-loading compartments.

(4) **Invariant Chain Degradation**

<FIGURE>

Acid-activated proteases progressively degrade the invariant chain, leaving the class II-associated invariant-chain peptide (CLIP) within the MHC class II groove.

(5) **CLIP Removal**

<FIGURE>

The nonclassical MHC class II molecule HLA-DM catalyzes CLIP release and edits the peptide repertoire by favoring ligands that form stable complexes with MHC class II.

(6) **Transportation to Exterior**

<FIGURE>

Stable peptide–MHC class II complexes are transported to the plasma membrane for recognition by $CD4^+$ T cells.

Core Concept 2.7: Exogenous CD1 MHC

(1) **Glycolipid Uptake**: Antigen-presenting cells internalize microbial or endogenous glycolipids into endosomal compartments.

(2) **Preparation and Expression of CD1**: CD1 molecules assemble in the RER with β2-microglobulin and traffic through the secretory pathway to the plasma membrane.

(3) **Internalization of CD1**: Surface CD1 molecules can be internalized and routed into endosomal compartments through signals in their cytoplasmic tails.

(4) **Glycolipid Loading**: Within the endosomal system, lipid-transfer proteins facilitate the loading of glycolipid antigens into the hydrophobic binding groove of CD1.

(5) **Transport to the Cell Surface**: Loaded glycolipid–CD1 complexes return to the plasma membrane for recognition by lipid-reactive T cells.

Several lymphocyte populations can recognize or respond to lipid antigens presented by members of the CD1 family.

- Some γδ T cells recognize lipid–CD1 complexes without requiring CD4 or CD8 co-receptors.
- Certain conventional αβ T cells, including $CD4^+$ populations, recognize antigens presented by CD1 molecules.
- Natural killer cells may respond indirectly to cytokines and cellular changes associated with CD1-restricted immune responses, although they do not use clonotypic TCRs to recognize lipid–CD1 complexes.
- Invariant natural killer T cells use a semi-invariant αβ TCR to recognize glycolipids presented specifically by CD1d.

3 T Cells

3.1 T Cell Receptor

3.1.1 TCR Complex Characteristics

<FIGURE>

Core Concept 3.1: αβ and γδ Receptors

<FIGURE>

Each mature T cell expresses either an αβ or a γδ T-cell receptor (TCR), both of which are heterodimers. Most circulating human T cells express the αβ form.

Both receptor classes belong to the immunoglobulin superfamily and resemble membrane-bound antigen-binding fragments. Each chain contributes three complementarity-determining regions (CDRs) to the antigen-recognition surface.

Basic residues within the TCR transmembrane segments form ionic interactions with acidic residues in the CD3 signaling subunits, stabilizing assembly of the complete receptor complex.

Unlike immunoglobulins, TCRs do not undergo somatic hypermutation or affinity maturation after activation. The CD4 or CD8 co-receptor stabilizes recognition of the appropriate MHC class and recruits signaling machinery to the engaged TCR.

αβ receptors undergo thymic selection to generate a repertoire capable of recognizing foreign peptide in the context of self MHC while remaining tolerant of self antigens.

The resulting receptor provides highly specific recognition of a peptide–MHC complex.

γδ receptors support a more innate-like mode of immune surveillance, although their antigen recognition remains clonally distributed and can be highly specific.

Many γδ T cells respond rapidly to conserved stress-associated or microbial ligands without conventional peptide–MHC restriction.

Depending on the subset, γδ T cells can recognize microbial metabolites, lipids, and stress-induced host molecules associated with bacterial or parasitic infection.

These cells are particularly enriched at mucosal and epithelial barriers, where they provide rapid tissue surveillance.

Core Concept 3.2: γε and δε Heterodimers

<FIGURE>

The CD3γ, CD3δ, and CD3ε chains possess extracellular immunoglobulin-superfamily domains and assemble as CD3γε and CD3δε heterodimers.

Each CD3 chain contains one immunoreceptor tyrosine-based activation motif (ITAM) within its cytoplasmic tail, allowing ligand engagement to be converted into an intracellular signal.

Core Concept 3.3: ζζ or ζη Heterodimers

<FIGURE>

The ζ and η chains do not contain extracellular immunoglobulin-superfamily domains.

They have extremely short extracellular segments and long cytoplasmic tails specialized for signaling.

Each ζ-family chain contains three ITAMs rather than the single ITAM present in each CD3γ, CD3δ, or CD3ε chain. The predominant TCR complex contains a ζζ homodimer, whereas ζη heterodimers are less common.

3.1.2 CD4 & CD8

Core Concept 3.4: CD4

<FIGURE>

CD4 is a co-receptor expressed by helper T cells and other subsets that binds nonpolymorphic regions of MHC class II during TCR recognition.

CD4 functions as a monomeric transmembrane glycoprotein.

Its extracellular region contains four immunoglobulin-superfamily domains.

The amino-terminal D1 domain contacts a nonpolymorphic surface formed principally by the membrane-proximal β2 domain of MHC class II, away from the peptide-binding groove.

The cytoplasmic tail associates noncovalently with the Src-family kinase Lck, thereby positioning Lck near the TCR–CD3 complex during antigen recognition.

Core Concept 3.5: CD8

<FIGURE>

CD8 is the characteristic co-receptor of cytotoxic T cells and binds a nonpolymorphic region of MHC class I.

It is expressed as a disulfide-linked CD8αβ heterodimer or CD8αα homodimer.

Each CD8 chain contributes one extracellular immunoglobulin-superfamily domain.

The amino-terminal domains bind primarily to the α3 domain of the MHC class I α chain, with additional contacts involving β2-microglobulin.

The cytoplasmic tail of CD8α associates with Lck, coupling co-receptor engagement to phosphorylation of the TCR signaling complex.

3.1.3 TCR Genes and Rearragement

Core Concept 3.6: Gene Sequence and Receptor Overview

<FIGURE>

(a) **α & δ Peptide**

<FIGURE>

The TCR α and δ loci occupy a shared region on the long arm of chromosome 14, with the δ locus nested within the α locus.

Like an immunoglobulin light-chain locus, the α locus lacks D segments and assembles a VJ exon from approximately 40 functional V segments, approximately 50 J segments, and one C region.

Like an immunoglobulin heavy-chain locus, the δ locus contains D segments and forms a V(D)J exon. It uses a limited set of V, D, and J segments followed by a single Cδ gene; some V segments are shared with the α locus.

(b) **β Peptide**

<FIGURE>

The TCR β locus lies on the long arm of chromosome 7.

Its organization resembles that of an immunoglobulin heavy-chain locus because it includes D segments.

Approximately 40 functional V segments precede two D–J–C clusters, whose J clusters contain six and seven functional segments, respectively.

(c) **γ Peptide**

<FIGURE>

The TCR γ locus lies on the short arm of chromosome 7.

Its organization resembles that of an immunoglobulin light-chain locus because it lacks D segments. A limited set of functional V segments rearranges with J segments associated with two principal Cγ regions.

Core Concept 3.7: α Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **δ Removal & VL-J Joining**

<FIGURE>

RAG1 and RAG2 recognize recombination signal sequences and mediate V–J joining through the same V(D)J recombination mechanism used at immunoglobulin loci. Imprecise junction formation, including P- and N-nucleotide addition, further increases receptor diversity. Rearrangement of the α locus deletes the intervening δ locus from that chromosome.

(3) **Result**

<FIGURE>

A completed α-chain locus contains a selected V segment joined to one J segment, followed downstream by unrearranged J segments, an intron, and the Cα region.

Transcription and RNA processing subsequently produce the mature α-chain message.

Core Concept 3.8: β Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **V–D–J Joining**

- *V–D–J rearrangement*

  <FIGURE>

  The TCR β locus rearranges sequentially: a D segment first joins a J segment, after which a V segment joins the resulting DJ unit.

  RAG-mediated cleavage at recombination signal sequences and imprecise end joining generate the rearranged VDJ exon. P- and N-nucleotide addition at both junctions substantially increases CDR3 diversity.

- *Direct V–J joining*

  <FIGURE>

  Direct V–J joining is not a normal productive pathway at the TCR β locus because the 12/23 recombination constraints enforce incorporation of a Dβ segment.

  Consequently, productive β-chain rearrangement uses V–D–J recombination rather than the alternative V–J scheme depicted in the rough notes.

(3) **Result**

- *V–D–J product*

  <FIGURE>

- *Noncanonical V–J product*

  <FIGURE>

A productive VDJ exon is subsequently transcribed and spliced to the appropriate Cβ region for β-chain expression.

Core Concept 3.9: γ Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **V–J Joining**

<FIGURE>

RAG1 and RAG2 mediate V–J recombination at the γ locus through recombination signal sequences. Junctional processing, including P- and N-nucleotide addition, diversifies the resulting CDR3 sequence.

(3) **Result**

<FIGURE>

The rearranged VJ exon is subsequently transcribed and spliced to a compatible Cγ region for receptor expression.

Core Concept 3.10: δ Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **V–D–J Joining and Result**

<FIGURE>

Productive TCR δ rearrangement generally joins a V segment to one or more D segments and then to a J segment, creating a V(D)J exon.

Unlike most antigen-receptor loci, the arrangement of recombination signal sequences at the δ locus can permit incorporation of multiple D segments into a single rearranged chain.

RAG-mediated recombination and junctional processing, including P- and N-nucleotide addition, generate extensive diversity within the δ-chain CDR3 region.

Transcription and RNA processing then join the rearranged variable exon to Cδ for expression of the mature δ chain.

Core Concept 3.11: Subsequent Expression

(1) The rearranged locus is transcribed into a primary RNA transcript.

TCR constant regions are encoded by exons that specify the extracellular constant domain, connecting peptide, transmembrane segment, and short cytoplasmic tail.

<FIGURE>

(2) The primary transcript receives a 5′ cap and a 3′ poly(A) tail.

(3) RNA splicing removes introns and excludes unused downstream gene segments, joining the rearranged variable exon to the appropriate constant-region exons.

<FIGURE>

(4) Translation of the amino-terminal leader sequence directs the ribosome to the RER, where the nascent receptor chain enters the secretory pathway and its transmembrane segment anchors it in the ER membrane.

(5) Signal peptidase removes the leader peptide during translocation into the RER.

(6) Folding, disulfide-bond formation, and initial glycosylation occur primarily in the RER, followed by additional glycan processing in the Golgi.

(7) Vesicles transport fully assembled TCR–CD3 complexes from the Golgi to the plasma membrane, where vesicle fusion displays the receptors on the cell surface.

Core Concept 3.12: History: ***Hendrick and Davis***

Mark Davis and Stephen Hedrick helped identify genes encoding the TCR through experiments designed to isolate T-cell-specific, somatically rearranged transcripts.

(1) **Selecting Membrane Bound Polysomes**

<FIGURE>

Hedrick and Davis enriched for transcripts encoding secreted or membrane proteins by isolating mRNA associated with membrane-bound polysomes.

In T cells, this fraction included transcripts encoding the TCR, CD3 subunits, and the CD4 or CD8 co-receptors.

In B cells, it included transcripts encoding membrane immunoglobulin and associated co-receptors.

Because T and B lymphocytes also express many shared membrane proteins, further subtraction was required to isolate T-cell-specific candidates.

(2) **Subtractive Hybridization**

<FIGURE>

During subtractive hybridization, B-cell-derived nucleic acids were used to remove cDNAs shared by the two lymphocyte lineages, enriching the remaining library for transcripts preferentially expressed by T cells.

(3) **Gene Comparison Across Multiple T Cells**

<FIGURE>

The enriched candidate pool contained approximately ten T-cell-specific cDNA clones, which were then screened for the somatic DNA rearrangement expected of an antigen-receptor gene.

Each candidate probe was compared across independently derived T-cell clones and a non-T-cell control by genomic blotting.

A candidate that remained in germline configuration produced the same restriction-fragment pattern across the different cell lineages.

By contrast, a rearranged candidate produced clone-specific restriction fragments because independent T cells had assembled different variable-region sequences.

Two candidates exhibited T-cell-specific rearrangement, enabling identification of genes encoding the TCR chains.

Alloreactivity is the ability of a T cell selected on self MHC to recognize a structurally distinct allogeneic MHC–peptide complex. This cross-reactivity underlies strong T-cell responses to genetically mismatched transplants (see the figure on MHC characteristics).

3.2 T Cell Development

Core Concept 3.13: Developmental Pathway

<TABLE>

3.3 TCR Antigen Response

Core Concept 3.14: Antigen Response Pathway

(1) **Antigen Recognition**

A professional antigen-presenting cell displays an antigen-derived peptide within the binding groove of an MHC class II molecule.

(2) **Binding**

A cognate $CD4^+$ helper T cell binds the peptide–MHC class II complex through its TCR, while CD4 stabilizes the interaction and recruits Lck to the CD3 signaling apparatus.

Superantigens are microbial proteins that crosslink particular TCR Vβ regions to MHC class II outside the conventional peptide-binding interface. This noncanonical interaction activates unusually large T-cell populations and can provoke life-threatening systemic cytokine release.

(3) **Signaling Cascade**

TCR engagement and co-stimulation initiate interconnected signaling cascades that reach the nucleus, alter gene expression, and commit the cell to activation, proliferation, and differentiation.

(4) **Transcriptional Activation**

- *Immediate Genes*

  Immediate signaling activates pre-existing transcription factors that coordinate the early inflammatory and proliferative response.

  Principal factors include NFAT, AP-1, and NF-κB.

- *Early Genes*

  Early-response genes encode cytokines, cytokine receptors, and regulatory proteins that reinforce activation and determine developmental fate.

  At this stage, naive helper T cells can differentiate into specialized subsets, including T_H1, T_H2, T_H17, and regulatory T cells. T_H17 cells should not be designated T_H3, a term historically used for a distinct regulatory phenotype.

  Cytokines produced by dendritic cells, macrophages, and other cells activate lineage-defining transcription factors in the responding T cell.

  The principal relationships between polarizing cytokines and lineage-defining transcription factors include the following:

  - IL-12 promotes T_H1 differentiation through T-bet;

- IL-4 promotes T_H2 differentiation through GATA3;
- TGF-β together with inflammatory cytokines such as IL-6 promotes T_H17 differentiation through RORγt; and
- TGF-β and IL-2 promote peripheral regulatory T-cell differentiation and FOXP3 expression.

- *Late Genes*

  Late-response genes encode effector cytokines, adhesion molecules, and other proteins required for migration and specialized immune function.

Core Concept 3.15: TCR Antigen Recognition to ZAP70

<FIGURE>

(1) An antigen-presenting cell displays a peptide–MHC class II complex to a cognate TCR.

(2) CD4 binds the MHC class II molecule, stabilizes the receptor complex, and positions its associated kinase Lck near CD3.

(3) Co-receptor-associated $p56^{Lck}$ phosphorylates tyrosines within CD3 and ζ-chain ITAMs, creating docking sites for ZAP-70.

(4) ZAP-70 binds doubly phosphorylated ITAMs through its tandem SH2 domains and is phosphorylated and activated by Lck.

(5) Activated ZAP-70 phosphorylates adaptor proteins, including LAT and SLP-76, which organize downstream pathways leading to transcriptional activation.
   - PLC-γ1 hydrolyzes PIP_2 into diacylglycerol (DAG) and inositol 1,4,5-trisphosphate (IP_3); IP_3 then stimulates Ca^{2+} release from the RER.
   - Ca^{2+} binds calmodulin and activates calcineurin, which dephosphorylates NFAT and permits its entry into the nucleus.
   - DAG activates protein kinase C-θ and contributes to activation of the canonical NF-κB pathway.
   - DAG-dependent signaling also activates the small GTPase Ras.
   - Ras activates mitogen-activated protein kinase cascades and AP-1, inducing genes required for proliferation and effector differentiation.

Co-stimulatory regulation determines whether TCR signaling produces productive activation or functional unresponsiveness.

The B7 ligands CD80 and CD86 are membrane proteins on antigen-presenting cells that bind either the activating receptor CD28 or the inhibitory receptor CTLA-4 on T cells.

CD28 engagement amplifies TCR signaling and promotes cell survival, metabolism, cytokine production, and proliferation.

CD28 co-stimulation increases transcription of IL-2 and expression of the high-affinity IL-2 receptor α chain (CD25), establishing an autocrine positive-feedback loop that supports clonal expansion. It does not generate a higher-affinity TCR α chain.

Recognition of antigen by a naive T cell in the absence of adequate CD28 co-stimulation can induce anergy, deletion, or another form of peripheral tolerance.

CTLA-4 engagement opposes CD28-mediated co-stimulation and limits the magnitude of T-cell activation.

CTLA-4 therefore restrains proliferation and effector development after activation.

Conventional resting T cells express little surface CTLA-4, but activation induces its expression and trafficking to the plasma membrane; regulatory T cells express CTLA-4 constitutively.

Core Concept 3.16: Lipid Rafts

During immunological-synapse formation, TCR microclusters and associated lipid-ordered membrane domains reorganize dynamically. Large phosphatases such as CD45 are initially excluded from close-contact signaling zones, while adhesion molecules and the cytoskeleton segregate receptors into central and peripheral supramolecular activation clusters; CD22 is a B-cell protein and does not organize the T-cell synapse.

(1) **Initial engagement**: TCR microclusters form rapidly at sites of peptide–MHC binding and recruit proximal signaling proteins.

<FIGURE>

(2) **Intermediate organization**: Actin-dependent transport moves signaling microclusters inward while integrins establish a surrounding adhesive zone.

<FIGURE>

(3) **Mature synapse**: The interface develops a bull's-eye organization, with a central supramolecular activation cluster surrounded by a peripheral adhesion-rich ring.

<FIGURE>

4 Cytokines and Signals

4.1 Receptor Signaling Types

Core Concept 4.1: Immunoglobulin

<FIGURE>

Structure: Immunoglobulin-superfamily receptors share extracellular domains built from antiparallel β sheets arranged in a characteristic immunoglobulin fold.

Individual family members combine these domains with distinct transmembrane and cytoplasmic signaling arrangements suited to their functions.

Ligands: Members of this broad family participate in antigen recognition, cell adhesion, and growth-factor signaling and therefore interact with diverse ligands rather than a single ligand class.

Core Concept 4.2: Type I Receptors

<FIGURE>

Structure: Type I cytokine receptors are single-pass membrane proteins that commonly assemble as heterodimers or higher-order complexes.

Several receptors share signaling subunits, such as the common β chain or common γ chain, allowing related cytokines to activate overlapping pathways.

Their extracellular cytokine-receptor homology domains are formed primarily from antiparallel β strands arranged in fibronectin type III-like folds.

The membrane-distal portion contains four conserved cysteine residues that form stabilizing disulfide bonds.

A conserved membrane-proximal WSXWS motif contributes to receptor folding, trafficking, and ligand-responsive organization.

Because these receptors lack intrinsic kinase activity, their cytoplasmic tails recruit Janus kinases (JAKs) that phosphorylate receptor-associated signaling proteins.

Ligands: Many type I cytokines are small monomeric proteins organized as four-α-helix bundles.

JAK–STAT pathway: Ligand-dependent receptor assembly initiates a direct route from the plasma membrane to transcriptional regulation.

(1) Ligand binding changes receptor geometry and juxtaposes receptor-associated JAKs, permitting their activation.

(2) Activated JAKs phosphorylate tyrosine residues on the receptor tails, creating docking sites for STAT proteins.

(3) Recruited STATs are phosphorylated, dimerize, enter the nucleus, and regulate expression of cytokine-responsive genes.

Core Concept 4.3: Type II Receptors

<FIGURE>

Structure: Type II cytokine receptors are single-pass membrane proteins that assemble as homo- or heterodimeric signaling complexes.

Their subunits possess cytoplasmic regions that associate with JAK-family kinases and recruit downstream signaling proteins.

The extracellular portions contain fibronectin type III-like cytokine-receptor domains.

Conserved cysteine residues form disulfide bonds that stabilize the extracellular structure, but type II receptors lack the canonical WSXWS motif of type I receptors.

Ligands: Type II receptors bind helical cytokines, principally interferons and members of the IL-10 family.

JAK–STAT pathway: Signal transduction follows the same general logic as in type I cytokine receptors.

Ligand binding reorganizes receptor-associated JAKs and promotes their activation.

The activated JAKs phosphorylate receptor tails to generate STAT docking sites.

Recruited STATs are phosphorylated, dimerize, and enter the nucleus to regulate target-gene transcription.

Mammals express four JAK proteins and seven STAT proteins. Individual receptor complexes recruit characteristic combinations of these signaling molecules, allowing different cytokines to produce distinct transcriptional programs.

Core Concept 4.4: Interleukin 17

<FIGURE>

Structure: IL-17 receptors constitute a distinct cytokine-receptor family whose subunits can form homo- or heteromeric complexes.

Ligand binding promotes higher-order receptor assembly, although the precise stoichiometry can vary among IL-17 family members.

The functional receptor for IL-17A and IL-17F contains IL-17RA and IL-17RC subunits; simplified trimeric descriptions should not be treated as a universal fixed stoichiometry.

Ligands: IL-17A is a disulfide-linked homodimeric cytokine with a predominantly β-sheet structure and potent pro-inflammatory activity.

Receptor engagement recruits the adaptor Act1 and activates NF-κB, MAPK, and C/EBP-family transcriptional pathways.

IL-17 is produced by T_H17 cells as well as several innate-like lymphocyte populations, including γδ T cells and some invariant natural killer T cells, and is particularly important for antimicrobial defense at epithelial barriers.

Core Concept 4.5: Tumor Necrosis Factor

<FIGURE>

Structure: Tumor necrosis factor receptors are type I transmembrane proteins characterized by extracellular cysteine-rich domains stabilized by disulfide bonds.

These repeated cysteine-rich modules form the elongated extracellular ligand-binding region.

Binding of trimeric TNF ligands promotes productive clustering and conformational rearrangement of receptor subunits.

Ligands: TNF-family ligands are trimeric, β-sheet-rich proteins that are commonly synthesized as membrane-bound juxtacrine signals and may subsequently be released in soluble form.

Receptor outcome depends on the family member and adaptor complex. TNFR1 can recruit TRADD and RIPK1 to activate canonical NF-κB and MAPK signaling, while alternative complexes involving FADD and caspase-8 can initiate apoptosis.

Core Concept 4.6: Chemokines

<FIGURE>

Structure: Chemokine receptors are seven-transmembrane G-protein-coupled receptors whose cytoplasmic regions interact with heterotrimeric G proteins.

Their seven membrane-spanning α helices transmit ligand-induced conformational changes to a G protein associated with the receptor's intracellular surface.

Chemokine recognition is frequently promiscuous: one receptor may bind several chemokines, and one chemokine may engage multiple receptors.

Ligands: Chemokines are small secreted proteins containing a flexible amino terminus, a three-stranded β sheet, and a carboxy-terminal α helix stabilized by conserved disulfide bonds.

G-protein pathway: Ligand engagement activates an associated heterotrimeric G protein.

The activated receptor acts as a guanine-nucleotide exchange factor, causing the Gα subunit to release GDP and bind GTP.

GTP-bound Gα separates functionally from the Gβγ dimer, and both components regulate downstream effectors controlling chemotaxis, adhesion, and cytoskeletal remodeling.

4.2 Immune Signaling Types & Themes

Core Concept 4.7: Immune Signaling Types

<TABLE>

Core Concept 4.8: Signaling Themes

<FIGURE>

(a) **Phosphorylation**

Reversible phosphorylation adds or removes phosphate groups from proteins or lipids, thereby altering their activity, localization, conformation, or molecular interactions.

This regulatory principle is central to BCR and TCR signaling, JAK–STAT pathways, G-protein-coupled receptor pathways, and the metabolism of membrane phosphoinositides.

(b) **Phospholipid Hydrolysis**

Phospholipid hydrolysis generates membrane-associated and soluble second messengers from phospholipid substrates.

A prominent example is the cleavage of phosphatidylinositol 4,5-bisphosphate (PIP_2) by phospholipase C.

This reaction produces two functionally distinct products:

- diacylglycerol (DAG), a nonpolar messenger that remains within the plasma membrane; and
- inositol 1,4,5-trisphosphate (IP_3), a soluble messenger that diffuses through the cytosol.

IP_3 binds its receptor on the RER and opens Ca^{2+} channels. The resulting rise in cytosolic Ca^{2+} activates calmodulin and calcineurin, which promote NFAT translocation into the nucleus.

(c) **Protein Hydrolysis**

Regulated proteolysis irreversibly cleaves a protein substrate and is therefore well suited to decisive signaling transitions. Restoration of the original state generally requires synthesis of a new intact protein.

Examples include proteolytic release of the Notch intracellular domain and caspase-mediated cleavage of cellular substrates during apoptosis.

(d) **Cytoskeletal Changes**

Cytoskeletal remodeling enables immune cells to polarize, migrate, adhere, phagocytose targets, and organize receptor signaling interfaces.

For example, chemokine engagement of a seven-transmembrane receptor activates heterotrimeric G proteins, induces inside-out integrin activation, recruits talin and other adaptor proteins, and couples adhesion receptors to actin remodeling and directed movement.

4.3 Cytokine Interaction Types

Core Concept 4.9: Cytokine Interaction Types

(a) **Pleiotropy**

<FIGURE>

Pleiotropy occurs when a single cytokine produces distinct effects in multiple target-cell populations.

For example, IL-4 coordinates a T_H2-associated response by acting on B cells, mast cells, macrophages, and helper T cells.

(b) **Redundancy**

<FIGURE>

Redundancy occurs when different cytokines can elicit overlapping biological outcomes.

For example, IL-2, IL-4, and IL-5 can each support aspects of activated B-cell proliferation under appropriate conditions.

(c) **Synergy**

<FIGURE>

Synergy occurs when two signals acting together generate a response greater than, or qualitatively different from, the response to either signal alone.

In a T_H2-associated response, IL-4 provides a principal signal for IgE class-switch recombination, while IL-5 can support the proliferation and differentiation of appropriately activated B cells.

(d) **Antagonism**

<FIGURE>

Antagonism occurs when one cytokine inhibits or counterbalances the effects of another.

For example, IFN-γ promotes T_H1-associated immunity and can oppose IL-4-driven T_H2 differentiation and IgE class switching.

(e) **Cascade Induction**

<FIGURE>

Cascade induction occurs when one cytokine stimulates a target cell to produce additional cytokines, thereby propagating and amplifying the response.

For example, IL-12 produced by activated macrophages and dendritic cells promotes T_H1 differentiation; T_H1-derived IFN-γ then enhances classical macrophage activation, reinforcing the cell-mediated immune response.